

NetworkPeer — Master End-to-End Testing & Operations Guide

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Platform Architecture, Feature Verification, Database Management & Deployment Manual

1. Executive Summary & Live Endpoints

NetworkPeer is a decentralized, cryptographically verifiable field-work coordination platform. It pairs clients posting on-site tasks with verified field workers who record tamper-proof photo, video, and GPS evidence. Escrow-backed payments release automatically upon client verification.

1.1 Live Production & Staging Architecture

Service / Layer	Deployment Target	Endpoint / Identifier	Status
Web Platform	Vercel (Edge /	https://networkpeer-platform.vercel.app	Live / Healthy

Service / Layer	Deployment Target	Endpoint / Identifier	Status
	Serverless SSR)		
Backend API (ALB)	AWS Application Load Balancer (eu-north-1)	http://networkpeer-staging-api-alb-969746120.eu-north-1.elb.amazonaws.com/api/v1	Live / Healthy
ECS Fargate	AWS Container Service (Task v6)	Port 3000 (Dockerized Node.js Fastify API)	Active (1/1 Tasks)
RDS PostgreSQL	AWS RDS PostgreSQL 16.15 with PostGIS	networkpeer-staging-postgis.c7e6uqswai46.eu-north-1.rds.amazonaws.com:5432	Active / 41 Migrations
Auth Broker	AWS Cognito User Pool	User Pool eu-north-1_GKLq1Fqzz , Client 45mgqp0jff5so65daesm3g4e1u	Active
SMS Lambda	AWS Lambda (eu-north-1)	networkpeer-staging-cognito-custom-auth (Node.js 20.x)	Active / 10s Timeout
Media Storage	AWS S3	s3://networkpeer-media-staging-trivaro (KMS Encrypted)	Active

2. Fast2SMS Integration & OTP Verification Guide

2.1 Fast2SMS Configuration State

Your Fast2SMS API key

(`mHrFcbV0L1XnCR6WUIAZ8hzJ3E25PKkQufo7Nl9Siywvgap0qYLPYc93ZohI0QWa82sGjqM1J5XAVFxB`) has been configured directly in the AWS Lambda trigger environment (`FAST2SMS_API_KEY`).

2.2 Unlocking Instant SMS Delivery in Fast2SMS Dashboard

Fast2SMS enforces a single initial website verification step before dispatching live OTPs to Indian carriers: 1. Log into your account at **fast2sms.com**. 2. Go to **OTP SMS** or **DEV API** in the navigation menu. 3. Click on **Website Verification / KYC**. 4. Enter your deployed platform URL: `https://networkpeer-platform.vercel.app` 5. Complete the Aadhaar/transaction verification if prompted. Once saved, Fast2SMS immediately routes all OTPs to physical phones without further code changes.

2.3 Real-Time OTP CloudWatch Fallback (Never Blocked)

Even while awaiting telecom carrier delivery, testing is **never blocked**. Every OTP generated by AWS Cognito is printed directly to CloudWatch logs in real time:

```
aws logs tail /aws/lambda/networkpeer-staging-cognito-custom-auth  
--region eu-north-1 --since 1m
```

Output:

```
INFO [AUTH_OTP] Phone: +919971536158 | Code: 630090
```

Enter this 6-digit code on the Web or Mobile app to immediately authenticate.

3. Web & Mobile App Testing Runbook

3.1 Web Client Sign-In Flow

1. Open <https://networkpeer-platform.vercel.app/auth> .
2. Notice the **+91 badge** is locked and non-editable.
3. Enter your 10-digit phone number (e.g. [9971536158](#)).
4. Click **Send Verification Code**.
5. When the 6-digit input appears, enter the code received on your phone or read from CloudWatch.
6. Click **Verify and Continue**. You are redirected into the Client Dashboard ([/client](#)).

3.2 Mobile Worker App Testing (Mac M1 via USB Cable)

1. Enable Developer Mode on Android Phone:

- Go to **Settings > About Phone** > tap **Build Number** 7 times.
- Go to **Settings > Developer Options** > enable **USB Debugging** and **Install via USB**.

2. Connect Phone to Mac M1:

- Connect via USB-C cable. On the phone, select **Always allow from this computer**.

3. Verify Device Connection:

```
adb devices
```

4. Reverse Local Ports:

```
adb reverse tcp:8081 tcp:8081  
adb reverse tcp:8787 tcp:8787
```

5. Launch Worker App:

```
cd apps/mobile
npm run android
```

6. Sign In:

- The worker app displays the **+91 badge**.
- Enter your 10-digit mobile number and submit.
- Input the verification OTP code.

4. End-to-End Feature Verification Lifecycle

4.1 Step 1: Post a Job (Client)

1. Go to **Client Dashboard > Post New Job** (`/client/jobs/new`).
2. Fill out the multi-step job wizard:
 - **Title & Description:** e.g., "Storefront Inspection & Signage Verification".
 - **Category:** Retail / On-Site Inspection.
 - **Location:** Click on the interactive map or enter coordinates.
 - **Budget:** Set task payment (e.g., ₹500).
 - **Required Evidence Subtasks:** Photo of entrance, Video of aisles, GPS location stamp.
3. Submit and confirm escrow funding. The job transitions from `FUNDING` to `POSTED` .

4.2 Step 2: Discover & Accept Job (Worker)

1. In the Android app, open the **Available Tasks** tab.
2. Nearby jobs appear sorted by proximity distance bands (`UNDER_1_KM` , `1_TO_5_KM` , etc.) without exposing private client info.
3. Tap the job to view details and tap **Accept Job**.
4. The job status transitions to `ASSIGNED` , and the exact address is unlocked for the worker.

4.3 Step 3: Capture & Upload Tamper-Proof Evidence

1. The worker arrives on site and taps **Start Task** (`IN_PROGRESS`).
2. Taps **Capture Evidence** for each subtask:
 - Takes a live camera photo or video.
 - The app stamps device EXIF data, timestamp, and high-accuracy GPS coordinates.
 - Computes SHA-256 hash of the media file locally.
 - Uploads directly to S3 via presigned POST credentials.
3. Taps **Submit for Review** (`SUBMITTED`).

4.4 Step 4: Review & Escrow Release

1. The client opens the **Review Studio** (`/client/review/:jobId`).
2. Inspects side-by-side evidence images/videos stamped with map coordinates.
3. Clicks **Approve Job**.
4. The backend database automatically transfers escrowed funds from the client's balance to the worker's wallet (`COMPLETED`).

5. Professional Database & Storage Management Guide

(Executive Runbook for Office & Leadership Presentations)

5.1 Connecting to RDS PostgreSQL via Desktop GUI (TablePlus / DBeaver)

For a professional presentation, use **TablePlus** (Mac-native, fast) or **DBeaver Enterprise**.

- **Host:** `networkpeer-staging-postgis.c7e6uqswai46.eu-north-1.rds.amazonaws.com`

- **Port:** 5432
- **User:** postgres (or your IAM database user)
- **Database:** networkpeer
- **Key Tables to Demonstrate:**
 1. `users` : Identity records, roles (`CLIENT` , `WORKER` , `ADMIN`), and phone numbers.
 2. `jobs` : Core job state machine (`POSTED` , `ASSIGNED` , `COMPLETED`), geometry points, and budgets.
 3. `job_subtasks` : Granular requirements per job.
 4. `job_subtask_media` : S3 keys, SHA-256 hashes, and verified GPS coordinates.
 5. `escrow_accounts` & `escrow_transactions` : Cryptographic financial ledger.
 6. `audit_logs` : Immutable security log of every state change.

Spatial Map Demonstration: In TablePlus or DBeaver, open the `jobs` table, select the `location` column, and toggle the **Map View**. It renders all job pins on an OpenStreetMap layer.

5.2 Admin Web Dashboard (`/admin`)

Open <https://networkpeer-platform.vercel.app/admin> : - **Live Job Kanban:** Real-time statuses across all active platform tasks. - **Worker & Client Roster:** Profile ratings, phone verifications, and KYC documents. - **Ledger Audit:** Escrow balance, dispute tracking, and platform transactions. - **Review Console:** Live media review interface.

5.3 S3 Evidence Storage Inspection

- **AWS Console:** Navigate to S3 > `networkpeer-media-staging-trivaro` > `/evidence/` .
- **Cyberduck / Transmit:** Connect via S3 access key to browse folders hierarchically by `{job_id}/{subtask_id}/` .

6. Running AWS Services & Monthly Cost Breakdown

AWS Resource	Configuration Tier	Staging Monthly Cost	Production (10k MAU)
ECS Fargate	1 Task (1 vCPU, 2 GB RAM)	\$15.00	\$60.00 – \$120.00
Application Load Balancer	1 ALB (eu-north-1)	\$22.00	\$28.00
RDS PostgreSQL (PostGIS)	db.t4g.micro / db.t4g.small (20 GB gp3)	\$18.00 – \$28.00	\$75.00 – \$150.00
AWS Cognito	User Pool (eu-north-1)	Free (< 50,000 MAU)	Free (< 50,000 MAU)
AWS Lambda	Custom Auth Trigger (128 MB)	Free (Free Tier)	< \$1.00
AWS S3	Evidence Bucket + KMS Encryption	< \$1.00	\$15.00 – \$40.00
Amazon CloudWatch	Logs & Metrics (5 GB ingested)	\$3.00 – \$5.00	\$10.00 – \$20.00
Fast2SMS	Transactional SMS (~₹0.15/SMS)	~\$2.00 – \$5.00	Usage-dependent

AWS Resource	Configuration Tier	Staging Monthly Cost	Production (10k MAU)
Estimated Total	Staging Cloud Footprint	~\$55 – \$75 / month	~\$190 – \$350 / month

7. Troubleshooting & Emergency Operations

1. OTP Not Arriving via SMS:

- Run `aws logs tail /aws/lambda/networkpeer-staging-cognito-custom-auth --region eu-north-1 --since 1m` to instantly get the code.
- Complete website verification in the Fast2SMS portal.

2. Restarting Backend API on ECS Fargate:

```
aws ecs update-service --cluster networkpeer-staging-cluster  
--service networkpeer-staging-api --force-new-  
deployment --region eu-north-1
```

3. Running Fresh Database Migrations:

```
cd NetworkPeer-main  
DATABASE_URL="postgres://postgres:<PASSWORD>@networkpeer-  
staging-postgis.c7e6uqswai46.eu-north-  
1.rds.amazonaws.com:5432/networkpeer" npm run migrate
```